CONFIDENTIAL ARCHITECTURE MEMORANDUM

Project: PrivateDocs-LLM Enclave Deployment

Date: October 2026 | Classification: Restricted

1. EXECUTIVE SUMMARY

This memorandum outlines the architectural standards for the PrivateDocs-LLM platform
The primary requirement is strict isolation: zero third-party API egress is permitted.
All prompt completions, vector embeddings, and chunk indexing must occur within
the private enclave perimeter.

2. COMPUTE HARDWARE AND BUDGETING

The primary production compute platform is the AMD Instinct MI300X with 192GB memo
The operational cost ceiling is established at .99 per hour for GPU compute,
with a reserve allocation of 40 hours preserved for final stress evaluations.
Local pre-flight smoke testing and offline verification must incur zero cloud spend.

CONFIDENTIAL ARCHITECTURE MEMORANDUM (Page 2

3. RETRIEVAL-AUGMENTED GENERATION SPECIFICATIONS
- Dense Vector Model: BAAI/bge-small-en-v1.5 running on CPU (dim=384)
- Sparse Lexical Engine: Rank-BM25 with token cleaning
- Fusion Method: Reciprocal Rank Fusion (RRF) with constant k=60
- Vector Database: Qdrant vector database (remote and persistent storage)

4. DATA INTEGRITY AND SECURITY AUDIT
All documents must support idempotent re-ingestion without duplicate chunk accumulat
Application loggers must enforce zero prompt logging: user questions and query content
must never be logged at INFO or WARN log levels.